



Shilpa

**Inclusive Mobile Learning App For Visually,
Hearing, Physically And Cognitively
Impaired Students (Grade 3-5)**

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Problems



- Digital learning tools used in Sri Lanka are not designed for inclusive primary education.

- Most applications:

Address only a single disability

Lack Sinhala-first support

Are not aligned with the Grade 3–5 national curriculum

- Students with disabilities rely heavily on external assistance to complete basic learning activities.
- Lack of interactive learning tools for disabled students





Research Gap

- Most educational apps support only one type of disability
- Limited integration of multiple assistive technologies in one system
- Lack of Sinhala-first inclusive learning platforms
- Limited support for Sri Lankan Sign Language and Braille learning
- Few systems provide adaptive learning based on student ability
- Limited support for alternative interaction methods (eye gaze, dwell time, voice) for physically disabled students





Main Objectives

- Develop an inclusive mobile learning application for Grade 3–5 students with disabilities.
- Integrate support for visual, hearing, physical, and cognitive impairments in one platform.
- Provide accessible learning interactions using voice, gaze, gesture, and assistive technologies.
- Deliver Sinhala-first curriculum-aligned learning content with interactive activities and assessments.





Solution Overview

Shilpa is a Sinhala-first inclusive mobile learning application designed for Grade 3–5 students with visual, hearing, physical, and cognitive impairments.

Functional Requirements

- Hearing impairment support: Sri Lankan Sign Language learning with real-time gesture recognition and feedback.
- Visual impairment support: Voice-guided lessons and automatic Braille quiz document generation and marks analyzing.
- Physical impairment support: Eye-gaze navigation, dwell-time selection, and voice command interaction.
- Cognitive learning support: AI-based adaptive learning activities that adjust to student performance.

Non-Functional Requirements

- Accessibility for different disabilities
- Usability with simple child-friendly interface
- Real-time performance
- System reliability for continuous learning activities





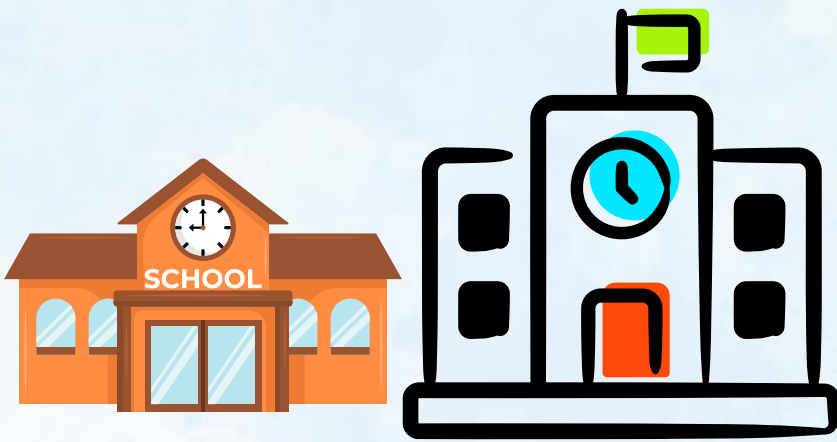
Target Users & Beneficiaries



- **Primary Users:** Grade 3–5 students with visual, hearing, physical, and cognitive disabilities



- **Secondary Users:** Teachers and parents for lesson personalization and progress tracking



- **Institutions / Society:** Schools, special education centers, and the Sri Lankan education system



- **Technology Need:** Inclusive, Sinhala-first, adaptive mobile learning platform



System Architecture & Integration

Architecture Diagram

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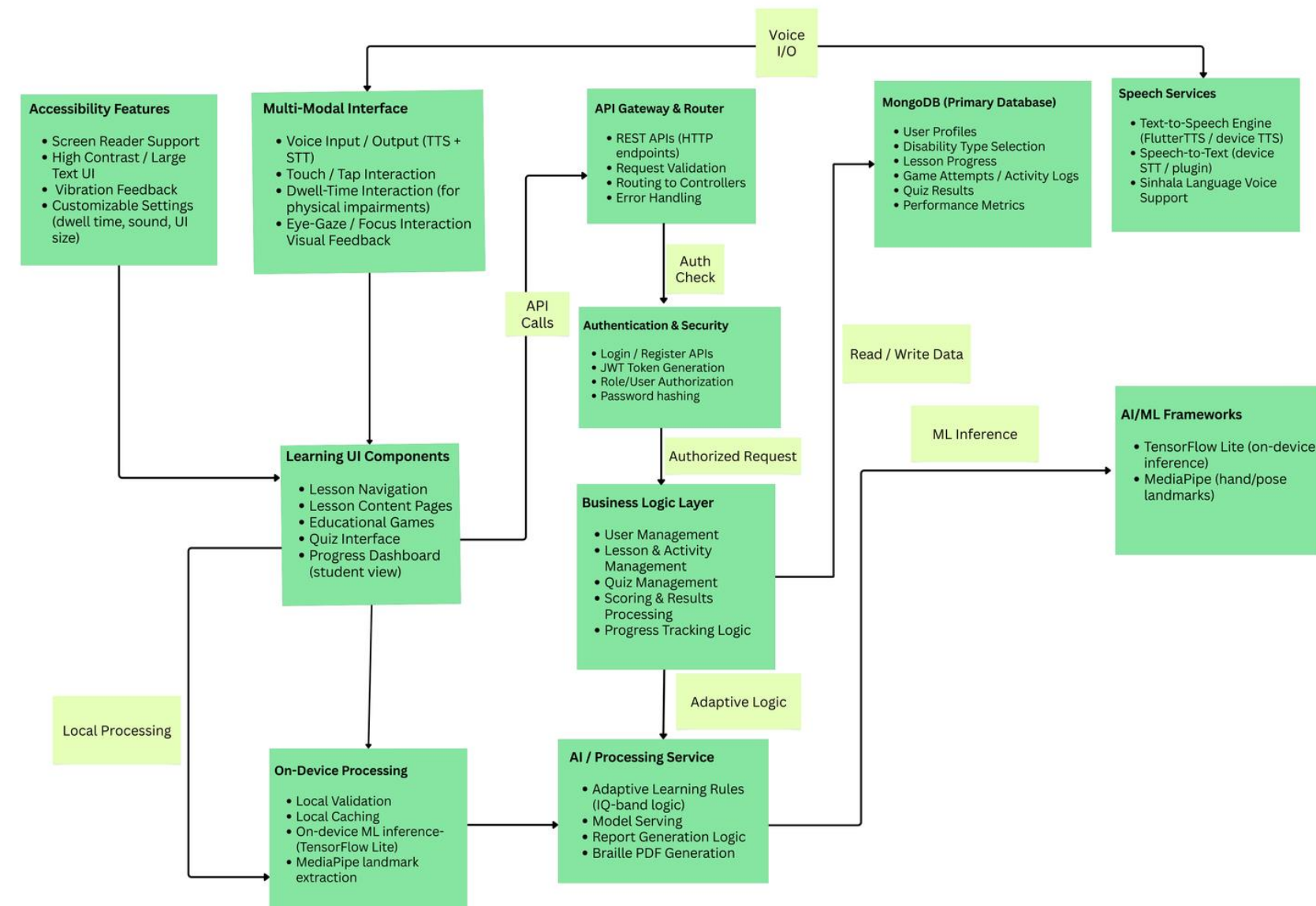


Frontend Services
(Flutter Mobile App)

Backend Services
(Node.js)

Data Layer
(Databases & Storage)

Integrations /
Third-Party Services





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Commercialization / Impact

Shilpa – Commercialization & Sustainability Customer Persona



Primary customers:

- Parents of Grade 3–5 students with disabilities
- Special education schools and inclusive primary schools

Secondary customers:

- Teachers, special educators, and NGOs
- Government and education-related institutions

Value proposition

- Single platform supporting visual, hearing, physical, and cognitive disabilities
- Sinhala-first curriculum aligned learning content
- Interactive lessons, quizzes, and progress tracking

Market Size

- A large number of primary school students with disabilities in Sri Lanka
- Growing demand for digital and inclusive education platforms

Adoption plan / next steps:

- Other grades (Grade 1–13)
- Other languages and regions

Cost and feasibility

- **Development cost:** Mainly labour for app development, AI model development, testing, and dataset preparation (sign language and Braille).
- **Operational cost:** Cloud hosting, database services, and periodic system maintenance/updates.
- **Feasibility:** Built with open-source technologies (Flutter, Node.js, TensorFlow Lite) and runs on standard Android devices with low infrastructure cost.





THANK YOU

